Practical No: 12

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Class:- CO3KB **Subject:-** CGR

Aim :- WAP for Line clipping using Cohen-Sutherland Algorithm

Program :-

```
#include <stdio.h>
#include <conio.h>
#include <graphics.h>
const int INSIDE = 0;
const int LEFT = 1;
const int RIGHT = 2;
const int BOTTOM = 4;
const int TOP = 8;
int x_min, y_min, x_max, y_max;
int computeCode(int x, int y) {
int code = INSIDE;
if (x < x_min)
code |= LEFT;
else if (x > x_max)
code |= RIGHT;
if (y < y_min)
code |= BOTTOM;
else if (y > y_max)
code |= TOP;
return code;
}
void cohenSutherlandClip(int x1, int y1, int x2, int y2) {
int code1 = computeCode(x1, y1);
int code2 = computeCode(x2, y2);
int accept = 0;

while (1) {
if ((code1 == 0) && (code2 == 0)) {
accept = 1;
break;
} else if (code1 & code2) {
```

```
break;
} else {
int code_out;
int x, y;
if (code1 != 0)
code_out = code1;
else
code_out = code2;
if (code_out & TOP) {
x = x1 + (x2 - x1) * (y_max - y1) / (y2 - y1);
y = y_max;
} else if (code_out & BOTTOM) {
x = x1 + (x2 - x1) * (y_min - y1) / (y2 - y1);
y = y_min;
} else if (code_out & RIGHT) {
y = y1 + (y2 - y1) * (x_max - x1) / (x2 - x1);
x = x_max;
} else if (code_out & LEFT) {
y = y1 + (y2 - y1) * (x_min - x1) / (x2 - x1);
x = x_min;
}
if (code_out == code1) {
x1 = x;
y1 = y;

code1 = computeCode(x1, y1);
} else {
x2 = x;
y2 = y;
code2 = computeCode(x2, y2);
}
}
}
if (accept) {
setcolor(WHITE);
line(x1, y1, x2, y2);
}
}
```

```
int main() {
int gd = DETECT, gm;
initgraph(&gd, &gm, "C:\\Turboc3\\BGI");
printf("Enter the clipping window coordinates (xmin, ymin, xmax, ymax): ");
scanf("%d %d %d %d", &x_min, &y_min, &x_max, &y_max);
rectangle(x_min, y_min, x_max, y_max);
int x1, y1, x2, y2;
printf("Enter the line coordinates (x1, y1, x2, y2): ");
scanf("%d %d %d %d", &x1, &y1, &x2, &y2);
setcolor(RED);
line(x1, y1, x2, y2);
cohenSutherlandClip(x1, y1, x2, y2);
getch();
closegraph();
return 0;
}
```

<u>OUTPUT</u> :-